END SEMESTER (REGULAR/RETEST) EXAMINATION, NOVEMBER/DECEMBER – 2026

Semester: 3rd
Branch: Mechanical
Subject Code: Me-302

FLUID MECHANICS AND HYDRAULIC MACHINES

Full Marks: 70
Time: Three hours

The figures in the margin indicate full marks for the questions.

Instructions:
(i) Objective question nos. 1, 2 and 3 are compulsory.
(ii) Answer question nos. 4 and 5 as per direction.

1. Fill in the blanks with approprite answers: [1x5=5]
   (a) The reciprocal of mass density is known as _____.
   (b) Crude oil is a _____ type of fluid.
   (c) The foot valve prevents the flow of liquid from the _____ to the _____ in a centrifugal pump.
   (d) Venturimeter are used to measure _____ through a pipe line.
   (e) Viscosity of liquid decreases with the _____ in temperature.

2. State whether the statements are True or False: [1x5=5]
   (a) A draft tube is used in reaction turbine.
   (b) The unit of density of fluid is $N/m^{3.}$
   (c) The specific gravity of Mercury is 13.6.
   (d) Chezy's formula is used to calculate major heada loss during the flow through pipes.
   (e) Hydraulic pump converts mechanical energy into hydraulic energy.

3. Choose the correct answers: [1x5=5]
   (a) In a reaction turbine, water at inlet possess
      (i) Only Kinetic Energy
      (ii) Only Pressure Energy
      (iii) Both (i) and (ii)
      (iv) None of these

   (b) Priming is done in order to
      (i) Run the pump satisfactorily
      (ii) Remove air from impeller and casing
      (iii) Completely fill the impeller and casing
      (iv) All of the above

(c) Loss due to friction in pipe is treated as
   (i) Minor loss
   (ii) Major loss
   (iii) Moderate loss
   (iv) Minimum loss

(d) Pitot tube is used to measure
   (i) Discharge
   (ii) Average velocity
   (iii) Velocity at a point
   (iv) Pressure at a point

(e) The value of normal atmospheric pressure is
   (i) 10.33 m of water
   (ii) 1.01325 bar
   (iii) 760 mm of Hg
   (iv) All of the above

4. Answer any five questions: [5x5=25]

(a) Define the terms: Specific gravity, Density, Viscosity

(b) If the specific gravity of petrol is 0.8, calculate the density, specific volume and specific weight.

(c) State and explain the Hydrostatic law.

(d) Define the terms: Total pressure and Meta center

(e) Differentiate between:
   (i) Laminar flow and Turbulent flow
   (ii) Steady and unsteady flow

(f) Define the three hydraulic co-efficient and derive the relation between them.

(g) State the Bernoulli's theorem. Write three assumptions made for Bernoulli's theorem.

(h) Define slip of a pump. What is negative slip? Why it occurs?

(i) Mention three differences between impluse turbine and reaction turbine.

(j) What is draft tube? What are its different types?

(k) What are the various losses of head (energy) in case of pipe flow?

5. Answer any five questions: [6x5=30]

(a) A plate 0.025 mm distant from a fixed plate, moves at 60 cm/sec and requires a shear stress of 2 N/m^2 to maintain this speed. Deterime the fluid viscosity between the plate the poise.

(b) Determine the total pressure and depth of pressure on a plane rectangular surface of 1 m wide and 3 m deep when its upper edge is horizontal and coincides with water surface.

(c) A differential manometer was connected with two pints at the same level in a pipe containing liquid of specific gravity of 0.85. Find the difference of pressure at the two points, if the distance of mercury levels be 150 mm.

(d) A horizontal venturimeter with inlet diameter 30 cm and throat diameter 15 cm is used to measure the flow of a oil with specific gravity 0.8. The discharge thorugh the venturimeter is 50 liter/sec. Find the readings of the oil mercury differential manomter. Take $C_d = 0.98$.

(e) Find the diameter of a pipe of length 2500 m when the rate of flow throught the pipe is 0.25 m^3/sec and the head lost due to friction is 5 m. Take C = 50 in Chezy's formula.

(f) A jet of water of diameter 50 mm strikes a fixed plate in such a way that the angle between the plate and the jet is 30°. The force exerted in the direction of jet is 1471.5 N. Deterimen the rate of flow of water.

(g) A pelton wheel is having a mean basket diameter of 0.8 m and is running at 1000 rpm. The net head of the pelton wheel is 400 m, if the side clearance angle is 15° and discharge through the nozzle is 150 liter/sec, find

(i) Power available at the nozzle.

(ii) Hydraulic efficiency of the turbine. Take $C_V = 0.98$.

COURSE OUTCOMES

Question no.	CO(s)
1. (a)	1
(b)	1
(c)	5
(d)	2
(e)	1
2. (a)	5
(b)	1
(c)	1
(d)	3
(e)	5
3. (a)	5
(b)	5
(c)	3
(d)	2
(e)	2
4. (a)	1
(b)	1
(c)	2
(d)	2
(e)	2
(f)	2
(g)	2
(h)	5
(i)	5
(j)	5
(k)	3
5. (a)	1
(b)	2
(c)	2
(d)	2
(e)	3
(f)	4
(g)	5